

Motion Planning and Simulation of a Wheel-Legged Robot Considering Transition Between Rolling and Walking Modes

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Abstract—The wheel-legged robot combines the rapid mobility of wheeled locomotion with the dexterous obstacle-crossing capability of legged movement. Optimizing its wheel-legged transition motion can further enhance its mobility, making rational planning of this transition particularly critical. In this study, kinematic analysis of the legged motion was first conducted to establish an inverse kinematic model, while a wheeled steering kinematic model was developed based on the Ackermann steering principle. Subsequently, characteristics of different gaits were analyzed, and gaits motion planning was implemented for the proposed robot model. Finally, foot placement timing and acceleration gain before/after ground contact during mode transition were optimized. Simulations on the ROS Gazebo platform demonstrate that this optimization can effectively reduce the average wheel motor load on support legs during transition, thereby improving the robot's terrain adaptability.

Keywords—wheel-legged robot; wheel-legged transition motion planning; gait planning; ROS simulation.

I. INTRODUCTION

With the advancement of modern society, the range of tasks undertaken by humans has continued to expand, encompassing activities such as search and rescue, border patrol, material handling, and field exploration. Many of these operational environments are inherently complex and variable, often containing unpredictable hazards, thereby necessitating the deployment of robots to perform such tasks in place of humans.

Wheeled robots possess high mobility speeds but exhibit weak obstacle-crossing capabilities. Legged robots possess the advantage of strong obstacle-crossing capabilities, and some current legged robots demonstrate excellent mobility on stairs or in other complex working environments [1], [2], but their movement speeds are relatively slow, making them unsuitable for some specific working environments. Wheel-legged robots not only possess the dexterous obstacle-crossing capabilities of legged robots [3] but also the high-speed mobility of wheeled robots [4], therefore they are widely used in applications such as military reconnaissance and extraterrestrial exploration [5], [6].

Yin et al. proposed an optimization framework named "Smart Gait" for hexapod robot gaits, which can significantly reduce energy consumption during robot motion [7]. Hu et al. proposed a gait planning method for hexapod robots called the Finite Incremental State Machine, greatly reducing the complexity of gait planning algorithms [8]. To facilitate user control of robots for various actions, Zhang et al. proposed an online gait transition algorithm based on foot position adjustment [9]. Qiu et al. proposed a hierarchical modular framework and learning details, using only a seven-dimensional vector to represent agent actions; the gait policies trained in their framework can generate new adaptive hexapod gaits on flat terrain [10]. Liu et al. proposed a fault-tolerant tripod gait planning method enabling stable walking with single-leg failure [11]. Cao et al. proposed a multi-algorithm collaborative control framework for multimodal coordinated operation, achieving efficient dynamic wheel-legged mode transition [12].

To reduce the average torque on the wheels of the support legs during wheel-legged transition in wheel-legged robots, thereby enhancing the robots' terrain adaptability, this paper proposes a method involving optimized foot placement timing and acceleration gain adjustment before and after foot placement. Simulation results demonstrate that the proposed method effectively reduces the average load on wheel motors, thereby enhancing the robot's operational capability in diverse environments. The main contributions of this paper are as follows: (1) Establish the kinematic equations of legged motion and wheeled steering motion for the wheel-legged robot. Analyze the characteristics of various legged gaits and perform motion planning for different gaits of the robot model in this paper. (2) Optimized the foot placement timing and acceleration gain adjustment before and after foot placement during the wheel-legged transition process. (3) Conducted simulation studies on different gaits and wheeled steering using the Gazebo platform in ROS, and performed comparative validation of the foot placement planning algorithm.

The paper is organized as follows: Section II presents the kinematic models for the robot's legged and wheeled

modes. Section III details legged gait planning based on the proposed model. Section IV optimizes foot placement timing and acceleration gain adjustment. Section V reports motion simulations and algorithm validation in the ROS environment. Section VI concludes the paper.

II. KINEMATIC MODEL OF THE ROBOT

The robot model adopted in this study is shown in Fig. 1, which is a six-wheel-legged robot.

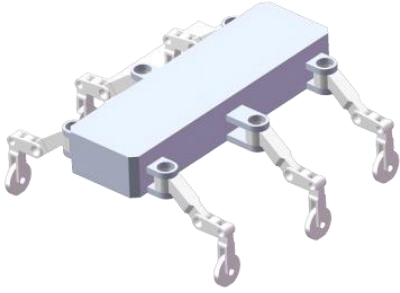


Figure 1. 3D model of the wheel-legged robot.

In coordinate system $\{O_A\}$, the position coordinates of the end-effector are denoted as ${}^A\mathbf{P} = ({}^Ax, {}^Ay, {}^Az)^T$; in coordinate system $\{O_L\}$, the position coordinates are denoted as ${}^L\mathbf{P} = (x_L, y_L, z_L)^T$. The transformation relationship between ${}^A\mathbf{P}$ and ${}^L\mathbf{P}$ can be expressed as

$$\begin{bmatrix} {}^L\mathbf{P} \\ 1 \end{bmatrix} = {}^L_A\mathbf{T} \begin{bmatrix} {}^A\mathbf{P} \\ 1 \end{bmatrix} \quad (1)$$

Based on (1), the inverse kinematics solution for a single leg of the robot is derived as (2). This equation enables the calculation of joint rotation angles from the end-effector's coordinates in the body reference frame for a given pose.

$$\begin{cases} \alpha = \tan^{-1}(x_L/y_L) \\ \beta = \arccos\left(\frac{K_1^2 + K_2^2 + L_2^2 - L_3^2}{2L_2\sqrt{K_1^2 + K_2^2}}\right) - \arctan\left(\frac{K_1}{K_2}\right) \\ \gamma = \arccos\left(\frac{K_1^2 + K_2^2 - L_2^2 - L_3^2}{2L_2L_3}\right) - \pi \\ K_1 = -z_L \\ K_2 = x_L/\cos\alpha - L_1 \end{cases} \quad (2)$$

Fig. 2 shows the geometric schematic of the inverse kinematics for a single robot leg, where α, β, γ represent the rotation angles of each joint, with the rotation direction coinciding with the Z-axis of each joint's reference frame.

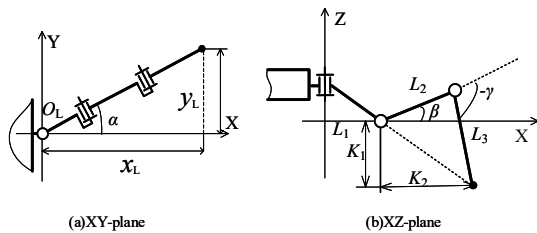


Figure 2. Single-leg inverse kinematics geometry.

Based on the Ackermann steering principle, kinematic modeling for the wheeled steering in this study is achieved. The principle formula for Ackermann steering is given by

$$\cot \delta_o - \cot \delta_i = B/L \quad (3)$$

where $\cot \delta_o$ denotes the steering angle of the outer front wheel, $\cot \delta_i$ denotes the steering angle of the inner front wheel, L represents the wheelbase, and B represents the track width.

Fig. 3 illustrates the Ackermann steering configuration adopted in this study. The distance from the centroid of the body to the steering center is the steering radius R of the Ackermann steering. During steering, the rotational speed of each wheel is proportional to the distance from its center to the steering center.

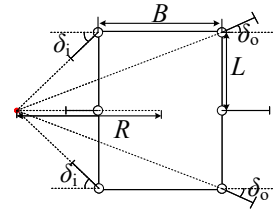


Figure 3. Schematic of the Ackermann steering.

According to Fig. 3, the steering angles in the diagram can be derived as

$$\begin{cases} \delta_i = \arctan\left(\frac{L}{R - \frac{B}{2}}\right) \\ \delta_o = \arctan\left(\frac{L}{R + \frac{B}{2}}\right) \end{cases} \quad (4)$$

III. GAITS PLANNING

For six-legged robots, three primary gaits enable stable locomotion through coordinated leg movements: The tripod gait involves alternating motion between two leg groups (three legs supporting/three legs swinging), forming a stable triangular support structure with strong terrain adaptability. The diagonal gait divides the six legs into three groups, maintaining two groups in ground contact and one group swinging. This gait balances stability and speed, suitable for uneven terrain. The wave gait operates each leg independently (five legs providing propulsion/one leg swinging), maximizing the support area for enhanced stability in complex terrain.

These gaits collectively facilitate stable walking through coordinated leg actions. The wave gait offers the highest stability, the tripod gait achieves the highest speed, and the diagonal gait provides intermediate performance.

In the six-wheel-legged robot, each individual leg has two states: swing phase and support phase. The state diagrams of legs under different gaits are shown in Fig. 4.

Define the left front leg as Leg 1, and number the legs counterclockwise starting from this leg. In the tripod gait, the leg movement of the robot is divided into two phases: the "support phase" and the "swing phase". In the tripod gait, the phase of each leg are as shown in Fig. 5: legs (1, 3, 5) propel the robot backward in support, while legs (2, 4, 6) lift off the ground and swing forward. legs (2, 4, 6)

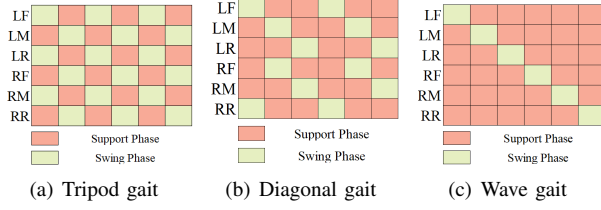


Figure 4. Gaits planning.

propel the robot backward in support, while legs (1, 3, 5) lift off the ground and swing forward.

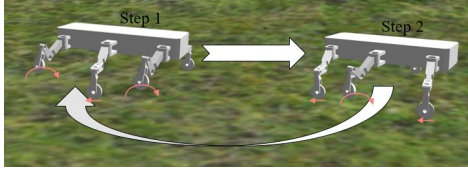


Figure 5. Tripod gait motion planning.

In the diagonal gait, the leg movement of the robot is divided into two phases: the “support phase” and the “swing phase”. In the diagonal gait, the phase of each leg are as shown in Fig. 6: legs (3, 5) lift off the ground and swing forward, while legs (1, 4) and legs (2, 6) propel backward in support. legs (1, 4) lift off the ground and swing forward, while legs (3, 5) and legs (2, 6) propel backward in support. legs (2, 6) lift off the ground and swing forward, while legs (1, 4) and legs (3, 5) propel backward in support.

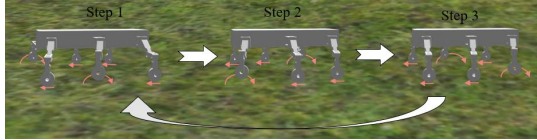


Figure 6. Diagonal gait motion planning.

In the wave gait, the leg movement of the robot is divided into two phases: the “support phase” and the “swing phase”. Leg 1 lifts off the ground and swings forward, while legs 2, 3, 4, 5, 6 propel backward in support. Leg 2 lifts off the ground and swings forward, while legs 1, 3, 4, 5, 6 propel backward in support. Leg 3 lifts off the ground and swings forward, while legs 1, 2, 4, 5, 6 propel backward in support. Leg 4 lifts off the ground and swings forward, while legs 1, 2, 3, 5, 6 propel backward in support. Leg 5 lifts off the ground and swings forward, while legs 1, 2, 3, 4, 6 propel backward in support. Leg 6 lifts off the ground and swings forward, while legs 1, 2, 3, 4, 5 propel backward in support.

IV. WHEEL-LEGGED TRANSITION MOTION PLANNING

During the wheel-legged transition process, the touchdown time of the swing leg and the acceleration profile of the wheel during transition both influence the average torque of the wheel motor. Reasonable planning of the touchdown time and the wheel's acceleration profile during transition

can effectively reduce the maximum value of the average torque, thereby enhancing the robot's terrain adaptability.

The robot's wheel-legged transition process is divided into two stages: the pre-touchdown stage and the post-touchdown stage. The wheel acceleration profile during the pre-touchdown stage is shown in (5), and the wheel acceleration profile during the post-touchdown stage is shown in (6). In these equations, both k_1 and k_2 are positive constants.

$$f(x) = k_1 \frac{A}{T} (1 - \cos(\frac{2\pi x}{T})) = k_1 a(x) \quad (5)$$

$$g(x) = k_2 \frac{A}{T} (1 - \cos(\frac{2\pi x}{T})) = k_2 a(x) \quad (6)$$

The total time required for the wheel-legged transition is T , and the touchdown time is t . The acceleration profiles before and after touchdown satisfy (7).

$$\int_0^t f(x)dx + \int_t^T g(x)dx = A \quad (7)$$

Simplifying (7) yields

$$\begin{cases} b(t)k_1 + (1 - b(t))k_2 = 1 \\ b(t) = \frac{t}{T} - \frac{1}{2\pi} \sin(\frac{2\pi t}{T}) \end{cases} \quad (8)$$

During the robot's wheel-legged transition, the average torque on the supporting wheel indirectly reflects the average load on the wheel motor at that moment. Considering the relatively small moment of inertia of the wheels in this paper's model, the average torque on the wheels before and after touchdown can be approximately represented by (9) and (10), respectively.

$$T_1 = \frac{f(x)mr^2}{n} \quad (9)$$

$$T_2 = \frac{g(x)mr^2}{N} \quad (10)$$

Consequently, the maximum value of the average torque on the supporting wheel during the entire wheel-legged transition process is given by (11).

$$P = \max \left\{ \frac{k_1 \max_{x \in [0, t]} a(x)mr^2}{n}, \frac{k_2 \max_{x \in [t, T]} a(x)mr^2}{N} \right\} \quad (11)$$

When the touchdown time is t , the objective function reaches its minimum value when the maximum values of the average torque before and after touchdown are equal. This is shown in (12).

$$\min P = \frac{k_1 a(t)mr^2}{n} = \frac{2k_2 A mr^2}{NT} \quad (12)$$

From (12), it follows that when $0 < t \leq \frac{T}{2}$, the relationship between k_1 , k_2 , t , and $\min P$ is given by (13).

$$\begin{cases} \min_{0 < t < \frac{T}{2}} P = \frac{2A \sin^2(\frac{\pi t}{T}) mr^2}{T [nb(t) + N \sin^2(\frac{\pi t}{T}) (1 - b(t))]} \\ k_1 = \frac{n \min P}{a(t)mr^2} \\ k_2 = \frac{NT \min P}{2A mr^2} \end{cases} \quad (13)$$

Similarly, when $\frac{T}{2} < t < T$, the relationship between k_1 , k_2 , t , and $\min P$ is given by (14).

$$\begin{cases} \min_{\frac{T}{2} < t \leq T} P = \frac{2A \sin^2\left(\frac{\pi t}{T}\right) m r^2}{T \left[n b(t) \sin^2\left(\frac{\pi t}{T}\right) + N(1 - b(t)) \right]} \\ k_1 = \frac{n T \min P}{2 A m r^2} \\ k_2 = \frac{\min P N}{a(t) m r^2} \end{cases} \quad (14)$$

It should be noted that the simulated annealing algorithm is employed to determine the minimum value of $\min P$ and the touchdown time t at which this minimum is achieved, thereby determining the values of the acceleration gains k_1 and k_2 before and after touchdown. The algorithm implementation is presented in **Algorithm 1**.

Algorithm 1: Wheel-legged motion transition optimization.

Input : T (total wheel acceleration time), A (max speed)
Output: t (foot contact time), k_1 (pre-contact acceleration gain), k_2 (post-contact acceleration gain)

```

1 while current_t > min_t do
2   for i = 1 to iter_per_temp do
3     Current_value = P(current_t),
4     if the Metropolis criterion is satisfied then
5       t = current_t,
6       min_P = Current_value,
7     end
8   end
9 end
10 Calculate  $k_1$  and  $k_2$  using (13) and (14),
11 return t,  $k_1$ ,  $k_2$ .
```

V. SIMULATION STUDY

The simulation platform was configured with an Intel(R) Core(TM) i5-8350U CPU @ 1.74 GHz and a 64-bit Ubuntu operating system, running the ROS/Gazebo environment. First, a realistic grassland scene was constructed and imported into Gazebo. Subsequently, a full-scale (1:1) model of the wheel-legged robot was designed in SolidWorks and integrated into Gazebo for simulation experiments.

Fig. 7 demonstrates the motion simulation of the wheel-legged robot using different gaits. During the simulation, the three gaits planned in this study could properly control the robot's locomotion.

Fig. 8 illustrates the simulation of the wheel-legged robot executing a steering maneuver based on the Ackermann steering principle.

To validate the accuracy of Ackermann steering, the robot's motion data was recorded in simulation and its trajectory deviation was calculated. Fig. 9 shows the comparison

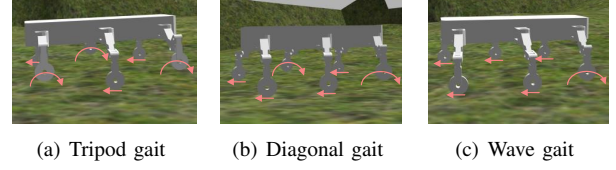


Figure 7. Three gaits simulation.

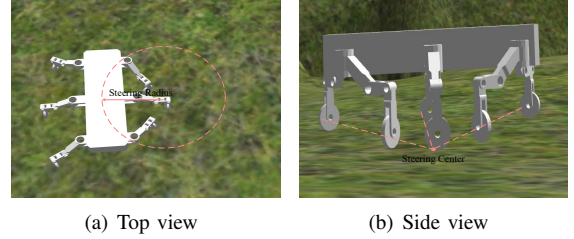


Figure 8. Steering simulation.

between the robot's actual motion trajectory and the desired trajectory, along with its motion error in the XY plane during movement. It can be seen from Fig. 9 that the maximum error during steering was less than 10 mm, which could basically meet the steering requirements of the wheel-legged robot.

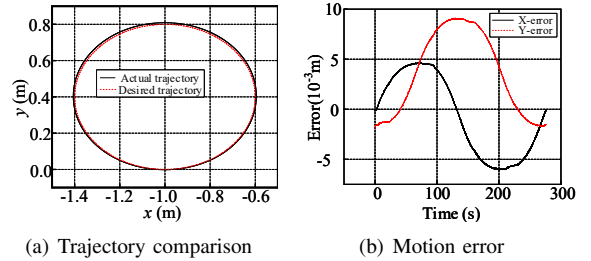


Figure 9. Steering simulation error.

Fig. 10 demonstrates the entire wheel-legged transition process of the robot. The entire wheel-legged transition process was divided into two stages: the pre-touchdown stage and the post-touchdown stage. Taking the tripod gait as an example: during the pre-touchdown stage, all six wheels maintained synchronized acceleration, while the three legs in swing phase continued to descend. During the post-touchdown stage, all six legs remained in support phase, and the six wheels maintained acceleration until reaching maximum speed.

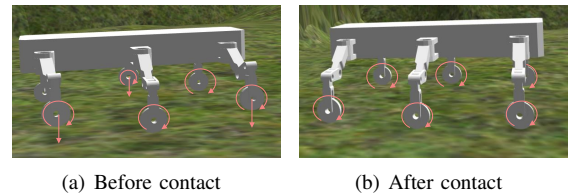


Figure 10. Snapshots of the wheel-legged transition process.

To test the effectiveness of **Algorithm 1** in wheel-legged transition optimization under three gaits, simulations were conducted under two conditions: baseline acceleration curves with optimized and non-optimized acceleration gains. The acceleration profile changes for the three gaits are shown in Fig. 11. According to Fig. 11, after optimization, the pre-touchdown acceleration gains for all three gaits were greater than those before optimization, while post-touchdown acceleration underwent a sudden change, with acceleration gains less than those before optimization.

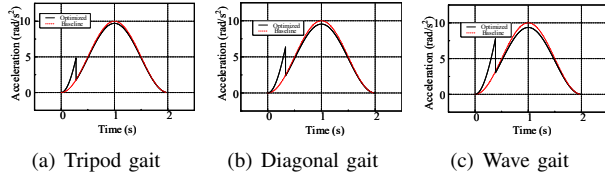


Figure 11. Acceleration of wheels.

To test the effectiveness of **Algorithm 1** in enhancing terrain adaptability across varying gaits, simulations were conducted under three conditions: unoptimized wheel-legged transition motion (Unoptimized), optimizing only the touchdown time (Time-Opt), and optimizing both the touchdown time and acceleration gains (Dual-Opt). The curves of average torque variation during the wheel-legged transition for the three gaits under the three conditions are shown in Fig. 12. According to Fig. 12, the robot's average torque underwent a sudden change at touchdown. After optimizing the touchdown time, the touchdown occurred earlier, effectively reducing the average torque. The maximum value of the curve optimized for both touchdown time and acceleration gains was smaller than the maxima of the other two curves.

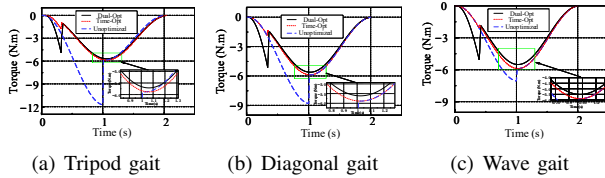


Figure 12. Average torque of support legs.

As seen in Fig. 12, optimizing the touchdown time and the pre- and post-touchdown acceleration gains could effectively reduce the maximum value of the average torque on the wheel during the wheel-legged transition process, thereby improving the robot's terrain adaptability.

VI. CONCLUSION

For a wheel-legged robot configuration with wheels at the foot ends, establish kinematic models for both legged and wheeled locomotion. Based on this model structure, motion planning for different gaits is accomplished. To enhance the robot's terrain adaptability, this paper proposes an algorithm to optimize the touchdown time and pre/post-touchdown acceleration gains during wheel-legged transition. The simulation results validate that the proposed algorithm effectively

reduces the average torque on the wheels during transition, thereby improving the robot's adaptability across diverse terrains. Our future work will focus on optimizing the transition trajectories of wheel-legged robots with diverse configurations.

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REFERENCES

- [1] G. Bellegarda, C. Nguyen, and Q. Nguyen, "Robust quadruped jumping via deep reinforcement learning," *Robotics and Autonomous Systems*, vol. 182, p. 104799, 2024.
- [2] H. Xing, Y. Xu, L. Ding, J. Chen, H. Gao, and M. Tavakoli, "Trajectory tracking control of wheeled mobile manipulators with joint flexibility via virtual decomposition approach," *IEEE Transactions on Automation Science and Engineering*, vol. 22, pp. 11 808–11 825, 2025.
- [3] M. Geilinger, R. Poranne, R. Desai, B. Thomaszewski, and S. Coros, "Skaterbots: Optimization-based design and motion synthesis for robotic creatures with legs and wheels," *ACM Transactions on Graphics (TOG)*, vol. 37, no. 4, pp. 1–12, 2018.
- [4] H. Xing, L. Ding, H. Gao, W. Li, and M. Tavakoli, "Dual-user haptic teleoperation of complementary motions of a redundant wheeled mobile manipulator considering task priority," *IEEE Transactions on Systems, Man, and Cybernetics: Systems*, vol. 52, no. 10, pp. 6283–6295, 2022.
- [5] Y. Wang, F. Pancheri, T. C. Lueth, and Y. Sun, "Design of a spider-inspired wheeled compliant leg for search mobile robots," *Biomimetic Intelligence and Robotics*, vol. 4, no. 4, p. 100182, 2024.
- [6] C. Liu, X. Zhang, R. Wu, S. Wang, R. Wang, Z. Lu, Y. Xiao, S. Li, and Z. Wu, "Mechanical design and wheel-leg-body cooperation control of a step-climbing robot," *Journal of Field Robotics*, vol. 39, no. 7, pp. 1054–1068, 2022.
- [7] Y. Yin, F. Gao, Q. Sun, Y. Zhao, and Y. Xiao, "Smart gait: a gait optimization framework for hexapod robots," *Chinese Journal of Mechanical Engineering*, vol. 37, no. 1, p. 15, 2024.
- [8] P. Hu, M. Zhang, and D. Wang, "Woa-fism planning hexapod robot various gaits," *Intelligent Service Robotics*, vol. 17, no. 4, pp. 963–979, 2024.
- [9] F. Zhang, S. Zhang, Q. Wang, Y. Yang, and B. Jin, "Straight gait research of a small electric hexapod robot," *Applied Sciences*, vol. 11, no. 8, p. 3714, 2021.
- [10] Z. Qiu, W. Wei, and X. Liu, "Adaptive gait generation for hexapod robots based on reinforcement learning and hierarchical framework," in *Actuators*, vol. 12, no. 2. MDPI, 2023, p. 75.
- [11] Y. Liu, X. Fan, L. Ding, J. Wang, T. Liu, and H. Gao, "Fault-tolerant tripod gait planning and verification of a hexapod robot," *Applied Sciences*, vol. 10, no. 8, p. 2959, 2020.
- [12] J. Cao, J. Zhang, T. Wang, J. Meng, S. Li, and M. Li, "Mechanism design and dynamic switching modal control of the wheel-legged separation quadruped robot," *Robotica*, vol. 42, no. 3, pp. 660–683, 2024.